



Incident report analysis

Summary	The company experienced a security event when all network services suddenly stopped responding. The cybersecurity team found the disruption was caused by a distributed denial of services (DDoS) attack through a flood of incoming ICMP packets. The team responded by blocking the attack and stopping all non-critical network services, so that critical network services could be restored.
Identify	A malicious actor or actors targeted the company with an ICMP flood attack. The entire internal network was affected. All critical network resources needed to be secured and restored to a functioning state.
Protect	The cybersecurity team implemented a new firewall rule to limit the rate of incoming ICMP packets and an IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics.
Detect	The cybersecurity team configured source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets and implemented network monitoring software to detect abnormal traffic patterns.
Respond	For future security events, the cybersecurity team will isolate affected systems to prevent further disruption to the network. They will attempt to restore any critical systems and services that were disrupted by the event. Then, the team will analyze network logs to check for suspicious and abnormal activity. The team will also report all incidents to upper management and appropriate legal authorities, if applicable.
Recover	To recover from a DDoS attack by ICMP flooding, access to network services need to be restored to a normal functioning state. In the future, external ICMP flood attacks can be blocked at the firewall. Then, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online.

Reflections/Notes: This activity strengthened my understanding of how the NIST Cybersecurity Framework (CSF) can be applied to real-world cybersecurity incidents. I learned how each function of the framework—Identify, Protect, Detect, Respond, and Recover—contributes to an organized incident response process. The exercise also highlighted the importance of proactive monitoring, layered network defenses, and having a well-defined incident response and recovery plan to minimize the impact of DDoS attacks.